Solutions Tutorial 2

Exercise 1

(a) We obtain the conditional factor demand function $z(w,q)$ through the cost minimisation problem:

$$\min_{z_1,z_2\geq 0} \; w_1 z_1 + w_2 z_2 \quad \text{subject to} \quad z_1^{\alpha} z_2^{\beta} \geq q.$$

We minimise the Lagrangian in the same way as for the EMP (assuming an interior solution):

$$\min_{z_1,z_2,\lambda} \; \mathcal{L} = w_1 z_1 + w_2 z_2 + \lambda\big(q - z_1^{\alpha} z_2^{\beta}\big).$$

$$\frac{\partial \mathcal{L}}{\partial z_1} = w_1 - \lambda\,\alpha\, z_1^{\alpha-1} z_2^{\beta} = 0 \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial z_2} = w_2 - \lambda\,\beta\, z_1^{\alpha} z_2^{\beta-1} = 0 \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q - z_1^{\alpha} z_2^{\beta} = 0 \tag{3}$$

From (1) and (2):

$$\frac{\alpha\, z_2}{\beta\, z_1} = \frac{w_1}{w_2} \iff z_2 = \frac{w_1}{w_2} \cdot \frac{\beta\, z_1}{\alpha}. \tag{$*$}$$

From (3):

$$q = z_1^{\alpha} z_2^{\beta} \iff z_1 = q^{\frac{1}{\alpha}}\, z_2^{-\frac{\beta}{\alpha}}. \tag{$**$}$$

Substituting $(**)$ in $(*)$:

$$z_2 = \frac{w_1}{w_2} \cdot \frac{\beta}{\alpha}\Big(q^{\frac{1}{\alpha}} z_2^{-\frac{\beta}{\alpha}}\Big) \iff z_2^{\frac{\alpha+\beta}{\alpha}} = \frac{w_1}{w_2} \cdot \frac{\beta}{\alpha}\, q^{\frac{1}{\alpha}}$$

$$\boxed{z_2(w,q) = \Big(\frac{w_1}{w_2}\Big)^{\frac{\alpha}{\alpha+\beta}} \Big(\frac{\beta}{\alpha}\Big)^{\frac{\alpha}{\alpha+\beta}} q^{\frac{1}{\alpha+\beta}}}$$

Similarly:

$$\boxed{z_1(w,q) = \Big(\frac{w_2}{w_1}\Big)^{\frac{\beta}{\alpha+\beta}} \Big(\frac{\alpha}{\beta}\Big)^{\frac{\beta}{\alpha+\beta}} q^{\frac{1}{\alpha+\beta}}}$$

(b) Cost function $c(w,q) = w_1 z_1(w,q) + w_2 z_2(w,q)$. Note that

$$w_1\Big(\frac{w_2}{w_1}\Big)^{\frac{\beta}{\alpha+\beta}} = w_2\Big(\frac{w_1}{w_2}\Big)^{\frac{\alpha}{\alpha+\beta}} = w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}},$$

so

$$\boxed{c(w,q) = w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} \underbrace{\left[\Big(\frac{\alpha}{\beta}\Big)^{\frac{\beta}{\alpha+\beta}} + \Big(\frac{\alpha}{\beta}\Big)^{-\frac{\alpha}{\alpha+\beta}}\right]}_{\equiv \Theta} q^{\frac{1}{\alpha+\beta}}}$$

Note: $c(w,q)$ is always homogeneous of degree one in w. Under CRS ($\alpha+\beta=1$), $c(w,q)$ is also homogeneous of degree one in q: $c(w,tq) = t\,c(w,q)$ for all $t>0$.

(c) The marginal cost of production is

$$\boxed{\frac{\partial c(w,q)}{\partial q} = w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} \,\Theta\, \frac{1}{\alpha+\beta}\, q^{\frac{1-(\alpha+\beta)}{\alpha+\beta}}}$$

Note: CRS $\Longrightarrow$ the marginal cost of production does not depend on q.

(d) The supply function is the netput vector

$$y(w,p) = \begin{pmatrix} y_1(w,p) \\ y_2(w,p) \\ y_3(w,p) \end{pmatrix},$$

where $y_3 = q$ is the output and $y_1 = -z_1$, $y_2 = -z_2$ are the (negative) inputs. We can reformulate the PMP as

$$\max_{q\geq 0}\; p\,q - c(w,q).$$

The necessary first-order condition for the profit-maximising q^* is

$$p - \frac{\partial c(w,q^*)}{\partial q} \leq 0 \quad \text{with equality if } q^*>0.$$

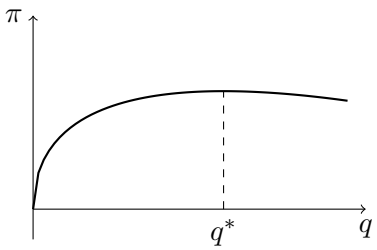
Given our answer to (c), we have

$$p \leq w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} \,\Theta\, \frac{1}{\alpha+\beta}\, q^{\frac{1-(\alpha+\beta)}{\alpha+\beta}}. \qquad (*)$$

- When $\alpha+\beta \leq 1$, this first-order condition is sufficient for a maximum; in this case the cost function is convex in q.
- When $\alpha+\beta < 1$ (decreasing returns to scale), $c(w,q)$ is strictly convex in $q \Longrightarrow$ unique solution. Solving $(*)$ with equality for q:

$$q(w,p) = \left(p \left[w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} \frac{\Theta}{\alpha+\beta} \right]^{-1} \right)^{\frac{\alpha+\beta}{1-(\alpha+\beta)}},$$

  and $y_3(w,p) = q(w,p)$ (output in our single-output setting).



$\alpha+\beta<1$: profit is concave in q, unique maximum

The unconditional factor demands $\tilde{z}_1(w,p)$ and $\tilde{z}_2(w,p)$ are obtained by substituting $q(w,p)$ in $z_1(w,q)$ and $z_2(w,q)$ of sub-question (a). So

$$y_1(w,p) = -\tilde{z}_1(w,p), \qquad y_2(w,p) = -\tilde{z}_2(w,p).$$

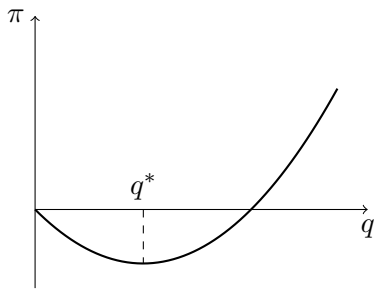
- When $\alpha + \beta = 1$ (CRS), the first-order condition equals

$$\underbrace{p}_{\text{market price of one unit of output } y_3} \leq \underbrace{\Theta\, w_1^{\alpha} w_2^{\beta}}_{\text{marginal cost of production}} .$$

(Under CRS, $\Theta = (\alpha/\beta)^{\beta} + (\beta/\alpha)^{\alpha} = 1/(\alpha^{\alpha}\beta^{\beta})$, so marginal cost equals $w_1^{\alpha} w_2^{\beta}/(\alpha^{\alpha}\beta^{\beta})$.)

  - If the marginal cost of production $> p \Longrightarrow q = 0$.
  - If the marginal cost $< p \Longrightarrow$ no solution exists; infinite profits can be obtained by increasing q.
  - If the marginal cost $= p \Longrightarrow$ any nonnegative output is a solution.

- When $\alpha + \beta > 1$ (increasing returns to scale), the first-order condition is not sufficient: $c(w,q)$ is strictly concave in q, so the point satisfying $(*)$ is a local minimum of profits $\Longrightarrow$ no solution (profits are unbounded).



$\alpha + \beta > 1$: profit is convex in q, local minimum

(e) The profit function $\pi(w,p)$ is

$$\boxed{\pi(w,p) = p\, q(w,p) - w_1 \tilde{z}_1(w,p) - w_2 \tilde{z}_2(w,p)}$$

Note: if $\alpha + \beta = 1$ and marginal cost $= p \Longrightarrow$ profits are equal to zero for any nonnegative output level.

Note (closed form for $\alpha+\beta < 1$): writing $c(w,q) = K q^{1/(\alpha+\beta)}$ with $K = w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} \Theta$, the first-order condition $p = \frac{K}{\alpha+\beta}\, q^{\frac{1-(\alpha+\beta)}{\alpha+\beta}}$ implies $c(w, q(w,p)) = (\alpha+\beta)\, p\, q(w,p)$, so that

$$\pi(w,p) = \big(1 - (\alpha+\beta)\big)\, p\, q(w,p) = \big(1 - (\alpha+\beta)\big)\, p \left(\frac{(\alpha+\beta)\, p}{K} \right)^{\frac{\alpha+\beta}{1-(\alpha+\beta)}} .$$

Profits are positive under decreasing returns to scale and vanish as $\alpha + \beta \to 1$.

Exercise 2

(a) *Homogeneity.* For $t > 0$, $\Phi(tw) = a_E(tw_E)^{1-\sigma} + a_O(tw_O)^{1-\sigma} = t^{1-\sigma}\Phi(w)$. Hence

$$c(tw, q) = q\left(t^{1-\sigma}\Phi(w)\right)^{\frac{1}{1-\sigma}} = t\, q\, \Phi(w)^{\frac{1}{1-\sigma}} = t\, c(w, q).$$

AC and MC. Write $\varphi(w) \equiv \Phi(w)^{1/(1-\sigma)} = c(w, 1)$, the cost of one unit of output. Then $c(w, q) = q\,\varphi(w)$, so

$$AC(q) = \frac{c(w, q)}{q} = \varphi(w), \qquad MC(q) = \frac{\partial c(w, q)}{\partial q} = \varphi(w).$$

Both equal the unit cost and are flat in q. Cost proportional to output means that doubling output exactly doubles the cheapest input bundle: the technology has *constant returns to scale.* (This is the degenerate case of the AC/MC picture: there is no efficient scale.)

(b) *Why Shephard's lemma holds* (the same argument as for $\nabla_p e = h$). Fix $\bar{w}$. The bundle $z(\bar{w}, q)$ produces q, so it remains feasible at any other input prices w, and the minimum cost can be no higher than its cost:

$$c(w, q) \le w \cdot z(\bar{w}, q) \quad \text{for all } w, \qquad \text{with equality at } w = \bar{w}.$$

So $\phi(w) = c(w, q) - w \cdot z(\bar{w}, q)$ is ≤ 0 everywhere and $= 0$ at $\bar{w}$: $\bar{w}$ maximises ϕ, and the first-order condition $\nabla\phi(\bar{w}) = 0$ gives $\nabla_w c(\bar{w}, q) = z(\bar{w}, q)$. The inequality also says that the fixed-mix cost $w \cdot z(\bar{w}, q)$ always weakly overstates true cost, which is what part (e) quantifies.

Applying it. $z_i(w, q) = \partial c(w, q)/\partial w_i$. By the chain rule,

$$z_E(w, q) = q \cdot \frac{1}{1-\sigma}\,\Phi(w)^{\frac{1}{1-\sigma}-1} \cdot (1-\sigma)\, a_E\, w_E^{-\sigma} = \boxed{q\, a_E\, w_E^{-\sigma}\, \Phi(w)^{\frac{\sigma}{1-\sigma}}}$$

(using $\frac{1}{1-\sigma} - 1 = \frac{\sigma}{1-\sigma}$), and by the same steps $z_O(w, q) = q\, a_O\, w_O^{-\sigma}\, \Phi(w)^{\frac{\sigma}{1-\sigma}}$.

Cost shares.

$$s_i(w) = \frac{w_i z_i}{c} = \frac{q\, a_i\, w_i^{1-\sigma}\, \Phi^{\frac{\sigma}{1-\sigma}}}{q\, \Phi^{\frac{1}{1-\sigma}}} = a_i\, w_i^{1-\sigma}\, \Phi^{\frac{\sigma-1}{1-\sigma}} = \frac{a_i\, w_i^{1-\sigma}}{\Phi(w)}.$$

The shares do not depend on q and sum to one, as they must.

Homogeneity of degree zero. Using $\Phi(tw) = t^{1-\sigma}\Phi(w)$,

$$z_E(tw, q) = q\, a_E\, t^{-\sigma} w_E^{-\sigma}\left(t^{1-\sigma}\Phi(w)\right)^{\frac{\sigma}{1-\sigma}} = t^{-\sigma}\, t^{\sigma}\, z_E(w, q) = z_E(w, q),$$

and likewise for z_O. (This is also what Shephard's lemma predicts in general: the derivative of a function that is h.d.1 in w is h.d.0 in w.)

(c) Take logs of z_E from (b):

$$\ln z_E = \ln q + \ln a_E - \sigma \ln w_E + \frac{\sigma}{1-\sigma}\ln \Phi(w).$$

We need the derivative of $\ln \Phi$ with respect to $\ln w_j$. Since $\partial\Phi/\partial w_j = (1-\sigma)\, a_j w_j^{-\sigma}$,

$$\frac{\partial \ln \Phi}{\partial \ln w_j} = \frac{w_j}{\Phi}\,\frac{\partial \Phi}{\partial w_j} = \frac{(1-\sigma)\, a_j w_j^{1-\sigma}}{\Phi} = (1-\sigma)\, s_j.$$

Hence

$$\frac{\partial \ln z_E}{\partial \ln w_E} = -\sigma + \frac{\sigma}{1-\sigma}(1-\sigma)\, s_E = -\sigma\,(1-s_E),$$
$$\frac{\partial \ln z_E}{\partial \ln w_O} = 0 + \frac{\sigma}{1-\sigma}(1-\sigma)\, s_O = \sigma\, s_O.$$

Why they sum to zero. Since $s_O = 1 - s_E$, the sum is $-\sigma s_O + \sigma s_O = 0$. This is not special to CES. Since z_E is h.d.0 in w (part (b)), $z_E(tw_E, tw_O, q) = z_E(w_E, w_O, q)$ for all $t > 0$. Differentiate both sides with respect to t and evaluate at $t = 1$; the right-hand side does not depend on t, so

$$w_E \frac{\partial z_E}{\partial w_E} + w_O \frac{\partial z_E}{\partial w_O} = 0.$$

Dividing by z_E turns each term into an elasticity, so the price elasticities sum to zero. Economically: if both input prices rise by the same percentage, relative prices are unchanged and the cost-minimising mix does not move. Only the *difference* $g_E - g_O$ moves input demand at given output. The own-price elasticity is negative, as concavity of c in w requires.

(d) *Cost.* From $c(w,q) = q\,\Phi(w)^{1/(1-\sigma)}$,

$$\frac{c(w^1,q)}{c(w^0,q)} = \left(\frac{\Phi(w^1)}{\Phi(w^0)}\right)^{\frac{1}{1-\sigma}}.$$

The trick is to write each term of $\Phi(w^1)$ relative to $\Phi(w^0)$:

$$\frac{\Phi(w^1)}{\Phi(w^0)} = \sum_{i\in\{E,O\}} \frac{a_i\,(w_i^1)^{1-\sigma}}{\Phi(w^0)} = \sum_i \underbrace{\frac{a_i\,(w_i^0)^{1-\sigma}}{\Phi(w^0)}}_{=\,s_i^0 \text{ by (b)}} \left(\frac{w_i^1}{w_i^0}\right)^{1-\sigma} = \sum_i s_i^0\,(1+g_i)^{1-\sigma}.$$

Raising to the power $1/(1-\sigma)$ gives the result. Note that the parameters a_E, a_O have disappeared: only the initial shares, the shocks and σ are needed.

Energy per unit of output. From (b), $z_E/q = a_E\, w_E^{-\sigma}\, \Phi^{\frac{\sigma}{1-\sigma}} = a_E\, w_E^{-\sigma}\, \varphi(w)^{\sigma}$, with $\varphi(w) = c(w,q)/q$. Taking the ratio at the two price vectors (a_E cancels):

$$\frac{z_E(w^1,q)}{z_E(w^0,q)} = \left(\frac{w_E^1}{w_E^0}\right)^{-\sigma} \left(\frac{\varphi(w^1)}{\varphi(w^0)}\right)^{\sigma} = (1+g_E)^{-\sigma} \left[\frac{c(w^1,q)}{c(w^0,q)}\right]^{\sigma}.$$

(e) With $s_E^0 = 0.30$, $s_O^0 = 0.70$, $\sigma = 0.4$, $g_E = 0.93$, $g_O = 0.059$:

(i) *Fixed mix.* If the firm keeps $z(w^0,q)$, each input's cost scales by its own price change, so

$$\frac{w^1 \cdot z(w^0,q)}{c(w^0,q)} - 1 = \sum_i s_i^0\, g_i = 0.30 \times 0.93 + 0.70 \times 0.059 = 0.279 + 0.041 = 32.0\%.$$

(ii) *Exact.* $1 - \sigma = 0.6$, so $1.93^{0.6} = 1.4837$ and $1.059^{0.6} = 1.0350$. Then

$$\frac{c(w^1,q)}{c(w^0,q)} = \big[0.30 \times 1.4837 + 0.70 \times 1.0350\big]^{1/0.6} = 1.1696^{1.667} = 1.298,$$

an increase of 29.8%.

(iii) *Energy per unit of output.* $(1.93)^{-0.4}\,(1.298)^{0.4} = 0.7687 \times 1.1100 = 0.853$: energy use per unit of output falls by 14.7%. (Other inputs per unit: $(1.059)^{-0.4}(1.298)^{0.4} = 1.085$, a rise of 8.5%.) The new energy share is

$$s_E^1 = \frac{0.30 \times 1.4837}{0.30 \times 1.4837 + 0.70 \times 1.0350} = \frac{0.4451}{1.1696} = 0.381.$$

The energy share *rises* from 30% to 38% even though energy use falls: with $\sigma < 1$, quantities respond less than proportionally to relative prices, so spending shifts towards the input that became dear.

Why (i) ≥ (ii) for any technology. The old bundle $z(w^0, q)$ still produces q, so it is a feasible choice in the cost minimisation problem at w^1. The minimum can be no larger than the cost of any feasible choice:

$$c(w^1, q) = \min_{z:\, f(z) \geq q} w^1 \cdot z \;\leq\; w^1 \cdot z(w^0, q).$$

This is exactly the inequality used to prove Shephard's lemma in (b): by that lemma, $w \mapsto w \cdot z(w^0, q)$ is the tangent plane of $c(\cdot, q)$ at w^0, and c lies below it (concavity of c in w). The gap ($32.0 - 29.8 = 2.2$ percentage points here) is the value of re-optimising the mix; it is zero only if the firm cannot substitute ($\sigma = 0$). The fixed-mix calculation overstates the cost of the shock, just as a fixed basket over-compensates a household.

(f) With CRS and price taking, an interior industry equilibrium requires $p = MC = AC = \varphi(w)$ (part (a)), so the output price moves one-for-one with unit cost:

$$\frac{p^1}{p^0} = \frac{\varphi(w^1)}{\varphi(w^0)} = 1.298 \quad (+29.8\%).$$

With constant-elasticity demand $Q = A\,p^{\varepsilon_D}$ and $\varepsilon_D = -1$:

$$\frac{Q^1}{Q^0} = \left(\frac{p^1}{p^0}\right)^{-1} = \frac{1}{1.298} = 0.770 \quad (-23.0\%).$$

Total energy use is output times energy per unit (CRS: $z_E \propto q$):

$$\frac{E^1}{E^0} = 0.770 \times 0.853 = 0.657 \quad (-34.3\%),$$

split into a mix effect (-14.7%, at fixed output) and a scale effect (output -23.0%).

Comparison with 2022. The *mix* effect is in line with the data: at fixed output this plant cuts energy by about 15%, the size of the observed cut in industrial gas use. The *scale* effect is not: the model predicts a 23% collapse in output, but output rose by about 5%. The model is partial equilibrium with a fixed demand curve; in 2022 industrial demand was shifting out (the post-COVID recovery), which more than offset the movement along the demand curve. Read through the model, the data say that essentially the whole gas cut came from substitution, not from scaling back output. Do not over-read the near-exact match of -14.7% and -15%, though: the EU figure aggregates firms with very different energy shares, and this calculation is for one energy-intensive plant.

Exercise 3

(a) We must check the two requirements of the definition, for every t.

Step 1: every observed plan lies in Y^O. Fix an observation s. By definition, $y^s \in Y^O$ if and only if $p^t \cdot y^s \leq p^t \cdot y^t$ for all t. This is exactly WAPM. So $y^s \in Y^O$ for every s.

Step 2: every observed plan is profit maximising in Y^O. Fix t and take any $y \in Y^O$. By the definition of Y^O, $p^t \cdot y \leq p^t \cdot y^t$. So no element of Y^O earns more than y^t at prices p^t.

Steps 1 and 2 together say that Y^O rationalises the data. □

Note where WAPM is used: only in Step 1. Step 2 holds by construction for any data. Without WAPM, Y^O would still be "maximised" by the observed plans, but some of those plans would not be in it.

(b) *(i) Closed.* Take $y^n \in Y^O$ for all n with $y^n \to y$. Fix t. For every n, $p^t \cdot y^n \leq p^t \cdot y^t$. The map $y \mapsto p^t \cdot y$ is continuous, so $p^t \cdot y^n \to p^t \cdot y$, and a limit of numbers that are all $\leq p^t \cdot y^t$ is itself $\leq p^t \cdot y^t$:

$$p^t \cdot y = \lim_{n \to \infty} p^t \cdot y^n \leq p^t \cdot y^t.$$

This holds for every t, so $y \in Y^O$.

(ii) Convex. Take $y, y' \in Y^O$ and $\lambda \in [0,1]$. For every t,

$$p^t \cdot \big(\lambda y + (1-\lambda)y'\big) = \lambda\, p^t \cdot y + (1-\lambda)\, p^t \cdot y' \leq \lambda\, p^t \cdot y^t + (1-\lambda)\, p^t \cdot y^t = p^t \cdot y^t,$$

so $\lambda y + (1-\lambda)y' \in Y^O$.

(iii) Free disposal. Take $y \in Y^O$ and $y' \leq y$. Since $p^t \gg 0$ and $y - y' \geq 0$, $p^t \cdot (y - y') \geq 0$. Hence for every t,

$$p^t \cdot y' \leq p^t \cdot y \leq p^t \cdot y^t,$$

so $y' \in Y^O$.

What has been shown. The lectures give necessity; (a) and (b) give sufficiency:

the data can be rationalised by some production set $\iff$ the data satisfy WAPM $\iff$ the data can be rationalised by a closed, convex production set with free disposal.

This is the firm-side counterpart of Afriat's theorem, and considerably simpler: no utility numbers have to be constructed, because profit is observable. *Implication for convexity:* any finite data set that is consistent with profit maximisation at all is also consistent with profit maximisation over a *convex* set. So finite price–quantity data can never reject convexity (or free disposal) separately from profit maximisation; these properties have no testable content of their own. (The same is true of convexity of preferences in Afriat's theorem.)

(c) Let Y rationalise the data and take any $y \in Y$. For each t, rationalisation says y^t earns at least as much as every element of Y at p^t, so $p^t \cdot y \leq p^t \cdot y^t$. This holds for all t, which is the definition of $y \in Y^O$. Hence $Y \subseteq Y^O$. □

So Y^O is the *largest* production set consistent with the data: an outer bound on the technology. Each observation contributes one half-space; the true technology must lie inside all of them.